

One Model to Dose Them All: A Population Pharmacokinetic Model for Multiple Aminoglycosides Across the Human Lifespan

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Dominic M. H. Tong, PhD, Jonathan D. Faldasz, PharmD, BCPS, Ron J. Keizer, PhD, PharmD, and Jasmine H. Hughes, PhD

Abstract

Aminoglycosides require therapeutic drug monitoring due to substantial pharmacokinetic variability across individuals. While population pharmacokinetic (popPK) models have been developed for individual drugs across age groups, and cross-drug maturation models have been explored, it is unclear whether pooling data across aminoglycosides improves predictive performance. We developed and evaluated a unified multi-aminoglycoside popPK model spanning neonates, children, and adults. De-identified data from 5659 amikacin, gentamicin, and tobramycin patients at 154 US institutions were split into training and testing datasets. Predictive performance in the test dataset was evaluated using mean percent error, normalized root mean square error (nRMSE), and clinical accuracy (trough concentrations correctly classified as below or above 1 mg/L and non-troughs within 20% of observation), and compared with selected published popPK models. To assess generalizability, the final model was refit using single-drug datasets and leave-one-subgroup-out datasets. The unified model achieved the lowest nRMSE across age groups, with absolute reductions of 0.6%-24.8%. Clinical accuracy was highest for the unified model, with *a posteriori* accuracy in pediatric patients of 56.4% versus 48% in the best published model. Leave-one-out refits yielded similar estimates but reduced precision, while single-drug models demonstrated substantially reduced precision, and one such model was not identifiable. This unified model improved predictive performance across drugs and age groups and demonstrated robustness to exclusion of individual drug-age group combinations. These findings support use of a unified aminoglycoside model for Bayesian model-informed precision dosing and demonstrate the advantages of pooling drugs within a class and across developmental stages.

Keywords

aminoglycosides, model-informed precision dosing, population pharmacokinetics, popPK models

Introduction

Aminoglycosides are an important first-line therapy for gram-negative infections.^{1,2} Their antibacterial efficacy is concentration-dependent, and dosing strategies are designed to achieve high peak concentrations for efficacy while permitting sufficiently low concentrations between doses to limit toxicity. However, aminoglycosides have a narrow therapeutic index, where insufficient exposure risks treatment failure and resistance, and overexposure increases the risk of ototoxicity and nephrotoxicity.³⁻⁷ Dosing is further complicated by substantial inter-individual variability in pharmacokinetics, driven by differences in renal function, body size, and developmental physiology.^{8,9} As a result, therapeutic drug monitoring (TDM) is recommended to individualize dosing and obtain optimal exposure.¹⁰ The integration of TDM with model-informed precision dosing (MIPD) enables Bayesian estimation of patient-specific exposure and supports data-driven dose adjustment.¹¹ The clinical performance of such approaches, however, depends on the availability of population pharmacokinetic (popPK) models that are both sufficiently predictive and generalizable to the treated population.¹²

Aminoglycosides share many physicochemical and PK characteristics: They are hydrophilic compounds that primarily distribute in the extracellular fluid, exhibit minimal protein binding, and are eliminated primarily by the kidneys.⁸ As a result, renal function, body size, and developmental age consistently emerge as the primary drivers of inter-individual variability in published popPK models.¹³⁻²⁰ Preliminary research suggests that models developed for one aminoglycoside can predict PK in patients receiving a different aminoglycoside better than models developed specifically for that drug.²¹ This suggests that a unified

InsightRX, San Francisco, CA, USA

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Corresponding Author:

Dominic M. H. Tong PhD, InsightRX, 548 Market St #88083, San Francisco, CA
Email: dominic@insight-rx.com

cross-drug modeling approach can improve predictive performance.

Numerous aminoglycoside popPK models have been developed, but most are drug-specific, age-restricted, and derived from single-center datasets. Such narrow development populations increase the risk of overfitting to local patient demographics, clinical practices, and assay methods, reducing the predictive performance of these models when applied to new institutions or age groups. Training on diverse, multi-center datasets could improve robustness and improve generalizability of these models. However, data availability differs across aminoglycosides. Amikacin is a last-line antibiotic, used for severe infections and drug-resistant tuberculosis,^{22–24} but its relatively limited use restricts the size of model development populations, whereas gentamicin represents a large and diverse source of PK data. Pooling data across aminoglycosides may therefore strengthen estimation of shared physiological relationships, improve the accuracy of parameter estimates, and enhance predictive performance across drugs and clinical settings.

Previous multi-drug modeling efforts within antimicrobials have demonstrated the feasibility of pooling data to identify shared physiological processes while retaining drug-specific PK characteristics.²⁵ However, the extent to which PK knowledge can be transferred across aminoglycosides and across the human lifespan remains uncertain. Here, we developed a multi-center popPK model integrating amikacin, gentamicin, and tobramycin across the human lifespan. We evaluated its suitability for MIPD across drugs and age groups by evaluating its predictive performance on a held-out test set. We also examined the extent to which pooling mitigates data sparsity by comparing performance of single-drug and pooled models, and by evaluating performance when an entire drug-age subgroup was excluded from model development. The latter scenario provides insight into the potential for knowledge transfer to populations not represented in the original model development population.

Methods

Patient Data Collection

Routine clinical care data for patients treated in the United States were entered into the InsightRX Nova precision dosing platform. These data were collected during routine health care operations at contracted health care entities and were de-identified prior to analysis. Because these data were not collected specifically for this study and the investigators had no access to subject identifiers linked to the data, the analysis did not meet the definition of human subjects research under 45 CFR 46.102(e). This determination was consistent with the NIH “Am I Doing Human Subjects Research?”

decision tool.²⁶ De-identified data were included in this analysis if the patient was treated with amikacin, tobramycin, or gentamicin, and had at least one recorded serum drug concentration. The lower limit of quantification of drug concentration varied between sites and ranged from 0.3 to 0.6 mg/L. Treatment courses were removed if concentration measurements were during or within 15 min of the end of infusion, or if they were missing infusion lengths, had dose amounts of 0, or had long gaps between doses. Samples taken more than 72 h after the previous dose, which is the expected window of sampling for routine clinical care, were also removed.

Weight, serum creatinine, age, height, and sex were collected for each patient. Clinician discretion was allowed for determination of patient sex, and practice regarding gender versus sex may have varied by institution. For patients younger than 2 years of age, gestational age was collected. For older patients, gestational age was assumed to be 40 weeks. Age group (neonate, child, adult) was determined from the patient’s date of birth entered at the time of dosing. Based on calculated age, the platform presents eligible age-group options, with predefined overlap at boundary ranges to allow clinical discretion (neonates: ≤ 15 weeks post-natal age; children: 10 weeks–21 years; adults: ≥ 18 years). Consequently, patients near category thresholds could be assigned to either adjacent age group.

Pharmacokinetic Modeling

Outlier Removal. Given the size of the dataset and its deidentified, retrospective nature, individual chart review for each patient was not possible; however, early exploratory analysis revealed a few likely erroneous data points (e.g., data entry issues). A semi-automated approach to outlier removal was therefore performed. An initial two-compartment popPK model with standard allometric scaling on all parameters, inter-individual variability on clearance and central volume of distribution, and combined proportional and additive residual error was used to iteratively remove outlier PK observations from this real-world clinical data. For each iteration, the model was fit to the full dataset in NONMEM, the observation with the highest conditional weighted residual (CWRES) was identified and removed, a revised dataset excluding that observation was saved, and the process repeated until up to 5% of observations had been removed. The number of observations to ultimately exclude to create the final dataset was determined through inspection of visual diagnostics (see Figure S1).

Model Development. Patients were randomly assigned to either a training dataset (70% of the data) or a testing dataset (30% of the data), as shown

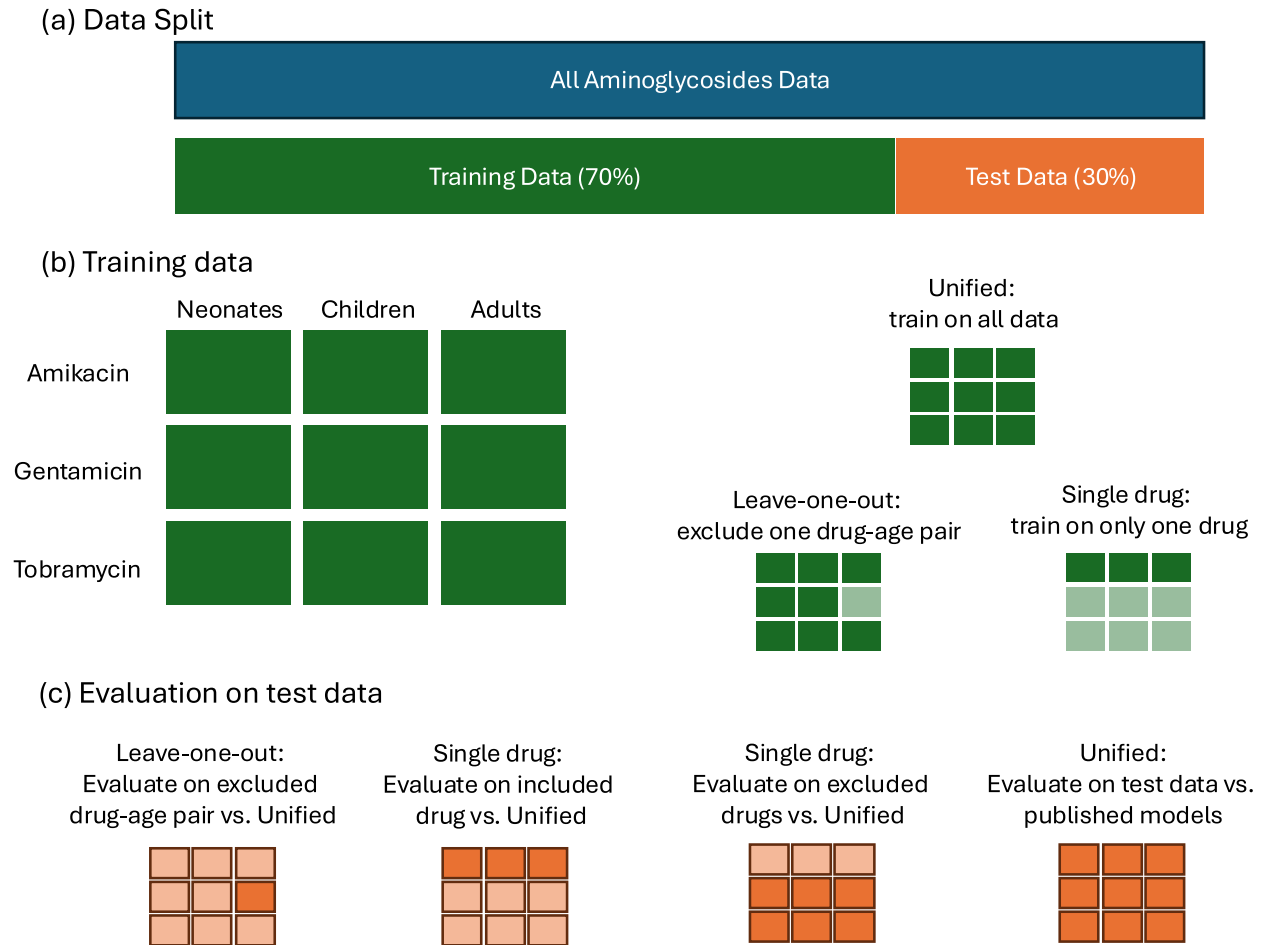


Figure 1. (a) Available data were split into 70% training and 30% test sets. (b) Training data were available across three drugs (amikacin, gentamicin, and tobramycin) and three age groups (neonates, children, and adults). The Unified model was trained on all available data; leave-one-out models were trained on all but one of the drug-age groups; and single drug models were trained only on one drug. (c) Leave-one-out models were evaluated against the Unified model on the excluded drug-age pair. Single drug models were evaluated against the Unified model on the single drug, and on the other two drugs. The Unified model was also evaluated against published models on the entire test dataset.

in Figure 1. Distributions of patient covariates were checked to confirm that training and testing datasets were comparable. Because two-compartment models are commonly used for aminoglycoside popPK modeling, particularly for amikacin and tobramycin, and have shown improved fit and predictive performance in several published analyses,^{15,27} model development focused on two-compartment structures. Improvements in model parameter precision, OFV, and goodness-of-fit plots were used to guide model development.

A base model without covariates was fitted first, after which all available covariates were evaluated using a manual stepwise covariate modeling approach. Covariates tested include postnatal age, postmenstrual age, chronological age, body weight, fat-free mass, serum creatinine, sex, height, cystic fibrosis status, and which drug was used. Serum creatinine and weight were implemented as time-varying covariates in the model.

Continuous covariates were modeled using a power model normalized to the median:

$$\left(\frac{\text{cov}}{\text{cov}_{\text{median}}} \right)^{\theta}$$

Discrete covariates were modeled using a power model:

$$\theta^{\text{cov}_{\text{discrete}}}$$

Since renal function varies across the human lifespan, several approaches were tested to describe the relationship between drug clearance, serum creatinine, age, and sex. Renal function with serum creatinine was modeled as an exponential model with the exponent as the difference from a typical serum creatinine^{28,29} for a patient with that age and sex:

$$\exp(\theta_{\text{CR}} \cdot (\text{SCR} - \text{SCR}_{\text{typical}}))$$

Renal maturation and age-related decay were modeled using sigmoidal functions, normalized to a 35-year-old patient²⁸:

$$\frac{\text{age}^\gamma}{\text{age}^\gamma + \text{EC}_{50}^\gamma}$$

We tested allometric scaling using total body weight across all patients. However, for hydrophilic drugs like aminoglycosides, fat-free mass (FFM) often better accounts for size-based PK parameter scaling than total body weight in adults.³⁰ As FFM for patients under 3 years of age is not well-defined, we defined a weight for allometric scaling using FFM (the M2 model by Al-Sallami et al³¹) down to 3 years old, then a linear interpolation of total body weight and FFM for patients between 1 and 3 years old, and then total body weight for patients below 1 year old, as follows:

$$W_{T_{\text{allo}}} = \begin{cases} WT, & \text{if } AGE < 1 \\ (-0.5 \cdot AGE + 1.5) \cdot WT + (0.5 \cdot AGE - 0.5) \cdot FFM, & \text{if } 1 \leq AGE < 3 \\ FFM, & \text{if } AGE \geq 3 \end{cases}$$

FFM is from the Al-Sallami model (M2), as follows:

$$\text{FFM} = \begin{cases} \left(1.11 + \frac{1-1.11}{1+(\frac{AGE}{7.1})^{-1.1}}\right) \cdot \left(\frac{9270 \cdot WT}{8780+244 \cdot BMI}\right) & \text{if female} \\ \left(0.88 + \frac{1-0.88}{1+(\frac{AGE}{13.4})^{-12.7}}\right) \cdot \left(\frac{9270 \cdot WT}{6680+216 \cdot BMI}\right) & \text{if male} \end{cases}$$

Inter-individual variability was evaluated on all pharmacokinetic parameters and was retained when it improved OFV, goodness-of-fit diagnostics, and overall model stability. Proportional and combined proportional and additive residual unexplained variability models were tested, and error magnitudes were estimated. Concentrations below the LOQ were handled using the likelihood (or M3) method for Bayesian estimation.^{32,33} All model fitting was performed using FOCE-I in NONMEM version 7.5.0 (ICON Development Solutions, Ellicott City, MD, USA), and subsequent data processing and analyses were conducted in R version 4.5.1.³⁴

Model Robustness and Generalizability Analyses

Two additional experiments were conducted using the training dataset to evaluate model robustness and generalizability (see Figure 1). In the first experiment, the final model structure was refit using data from only one aminoglycoside at a time. Parameters that were not estimable (e.g., CF effect on clearance in datasets without tobramycin patients, or the effect of amikacin on Q and V2 in datasets without amikacin patients) were

fixed. Predictive performance was evaluated separately on test data from the included drug and on test data from the two excluded drugs, allowing assessment of within-drug performance as well as cross-drug generalizability. In the second experiment, the final model was refitted on the training data with one drug-age group pair excluded at a time (e.g., excluding neonatal gentamicin data). Predictive performance was then evaluated on the excluded drug-age group pair to assess model performance on an unseen patient population. Each experiment was also evaluated based on its impact on parameter estimates, its influence on the age-clearance relationship, and its predictive performance.

Comparison With Published Population Pharmacokinetic Models

Traditional evaluation of popPK models often relies on simulation-based diagnostics such as visual predic-

tive checks (VPCs) and other goodness-of-fit analyses. However, these approaches do not fully capture how models are used in clinical practice, where dosing decisions are made iteratively based on limited data available at the time of dosing. To better reflect real-world clinical decision-making, we evaluated the ability of each model to predict future drug concentrations iteratively. That is, only data that were available at the time of clinical decision-making, that is, following collection of a drug level or a pair of drug levels, were used to inform Bayesian estimation of PK parameters, and these parameters were used to predict future drug levels. Drug concentrations that were likely interpreted as a set were grouped. This included two concentrations within the same dosing interval, usually a peak concentration and a trough concentration, or a trough concentration, followed by a dose, followed by a peak concentration.

Model predictive performance was compared with a selected subset of published popPK models in patients receiving aminoglycosides in the context of MIPD. Because the goal of this project was to assess the feasibility of fitting a model across multiple drugs of a similar class, a comprehensive literature review of models was not performed. Instead, a limited number of models for each drug and for each age group were selected based on their demonstrated performance in adult, pediatric, or neonatal populations, their use of commonly collected covariates, and their existing implementation in a Bayesian MIPD software tool (InsightRX Nova). For

fair comparison, each published model was evaluated only within the age group for which it was originally developed. This restriction was also necessary because several commonly used covariates, such as creatinine clearance as estimated by Cockcroft–Gault³⁵ and FFM estimated by Janmahasatian et al,³⁰ are not defined across all age groups. Model predictive performance was also evaluated across BMI classes (NIH standard classifications), biological sex, age, and renal function (<30, 30–60, 60–90, and >90 mL/min/1.73m², estimated by CKD-EPI for adults and the modified Schwartz equation for children and neonates; groups were consolidated if there were fewer than 30 samples in each group).

Statistical Analysis

The predictive performance of each model was evaluated using mean percent error (MPE) and normalized root mean square error (nRMSE) using the *i*-th prediction ($\hat{y}_i$) and observation (y_i), as follows:

$$\text{MPE} = \frac{1}{N} \sum_{i=1}^N \frac{\hat{y}_i - y_i}{y_i}$$

$$\text{nRMSE} = \frac{\sum_{i=1}^N y_i}{N} \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2}$$

RMSE was normalized to facilitate comparison across drugs, as amikacin requires higher concentrations for efficacy compared to gentamicin and tobramycin.

Clinical accuracy was also evaluated. For trough samples, which were taken within the last 20% of the dosing interval, a prediction was deemed accurate if the predicted and measured value were both above 1 mg/L or below 1 mg/L. For all other samples, a prediction was considered clinically accurate if the prediction was within 20% of the measured value. Uncertainty in these error metrics was assessed by bootstrapping 1000 samples and computing the error metric across these samples. Statistical significance was determined by the overlap of the 5th to 95th percentiles of bootstrapped samples of performance parameters. Model predictive performance evaluation was conducted using the R package *mipdeval* version 0.1.0.9000 in R version 4.5.1.

Results

Patients and Data Collection

A total of 1605 treatment courses were removed due to concentration measurements during or within 15 min of the end of infusion ($N = 692$), or due to assumed inaccurate dosing records (missing infusion lengths, dose amounts of 0, or long gaps between doses; $N =$

913). In addition, 251 observations were removed as they were taken more than 72 h after the previous dose, which could indicate missing dose records since routine samples are usually not collected at such extended time points.

The semi-automated outlier removal process removed another 150 observations (1.1% of observations), based on the parameter estimates plateauing and changes in objective function value (see Figure S1).

The final dataset after all data exclusions included 5658 patients and 5972 treatment courses (3235 adults, 510 children, and 2227 neonates) treated at 154 hospitals in the United States between January 2013 and December 2025 and summarized in Table 1. These treatment courses provided 13,822 serum drug concentrations (1865 amikacin, 7984 gentamicin, and 3973 tobramycin), of which 647 (4.7%) were below LOQ.

Population Pharmacokinetic Model

Table 2 summarizes the parameter estimates for the unified aminoglycoside popPK model. The two-compartment model adequately described the observed data across drugs and age groups. Inter-individual variability was included on clearance and central volume. Residual unexplained variability was best described by a combined proportional and additive error, with error magnitude estimated separately for each drug. All parameters were estimated with acceptable precision (RSE < 25.0%), except maturation T50 (59.3%) and additive error magnitude for tobramycin (75.3%).

Body size, renal function, cystic fibrosis status, drug, and age-related maturation and decay were identified as significant covariates on clearance. Allometric scaling based on WT_{allo} , with exponents fixed at 0.75 for clearance and inter-compartmental clearance and 1 for central and peripheral volumes, improved model fit relative to both the base model without allometric scaling and a model scaled using total body weight. Renal function was best described using deviation from age- and sex-adjusted expected serum creatinine, standardized by age and sex. Both renal maturation and age-related decline in clearance were modeled using sigmoidal functions, similar to the approach taken by Colin.²⁸ The resulting effect sizes were similar to Colin, though with a slightly later maturation timing (see Figure S2).

Clearance of tobramycin and gentamicin was 17% and 11% higher compared to amikacin, respectively. Presence of CF, where all patients were treated with tobramycin, increased clearance by 6%. Peripheral volume for amikacin was estimated to be 41% higher relative to the other two aminoglycosides, while intercompartmental clearance was estimated to be 16% lower ($p < 0.05$). In contrast, central volume of distribution did not differ significantly across aminoglycosides.

Table 1. Patient Characteristics of the Overall Dataset, the Training Dataset for Model Fitting, and the Validation Dataset for Model Evaluation

Characteristic	All	Training Set	Test Set
# of Patients	5972	4179	1793
Age group (adult/child/neonates)	3235/510/2227	2263/356/1560	972/154/667
Adult patients (ami/genta/tobra)	409/1813/1013	286/1268/709	123/545/304
Pediatric patients (ami/genta/tobra)	39/171/300	27/119/210	12/52/90
Neonatal patients (ami/genta/tobra)	104/1988/135	73/1392/95	31/596/40
Drug (ami/genta/tobra)	552/3972/1448	386/2779/1014	166/1193/434
Sex (female/male)	3169/2803	2219/1960	950/843
# of TDMs (ami/genta/tobra)	13822 (1865/7984/3973)	9681 (1341/5553/2787)	4141 (524/2431/1186)
Cystic fibrosis (absent/present)	5898/74	4125/54	1773/20
Age (years)	22 (0-100)	22 (0-97)	22 (0-100)
BMI (adults only; kg/m ²)	28 (11-79)	29 (11-73)	28 (12-79)
Baseline serum creatinine (mg/dL)*	0.70 (0.07-13)	0.70 (0.07-11)	0.70 (0.07-13)
Serum creatinine (mg/dL)	0.70 (0.06-13)	0.70 (0.06-11)	0.70 (0.07-13)
Baseline CKDEPI (mL/min/1.73m ²)*	100 (2.7-190)	100 (2.7-190)	100 (4.1-180)
CKDEPI (mL/min/1.73m ²)	95 (2.7-190)	96 (2.7-190)	95 (3.6-190)
Baseline Cockcroft–Gault (mL/min)*	33 (4.1-540)	32 (6.1-540)	33 (4.1-470)
Cockcroft–Gault (mL/min)	37 (4.1-540)	34 (6.1-540)	38 (4.1-470)
Fat-free mass (kg)	38 (0.27-120)	38 (0.30-120)	39 (0.30-120)
Height (cm)	150 (20-210)	150 (21-210)	160 (20-200)
Weight (kg)	55 (0.3-200)	55 (0.3-200)	55 (0.3-200)

Values are count or median (range). Baseline values (*) refer to the value that immediately preceded the first dose if available; otherwise, the first value within 48 h of dosing, else the median of all available values.

Diagnostic plots (see Figures S3–S10) demonstrated good agreement between observed and predicted concentrations across drugs, age groups, BMI groups, renal function, sex, and time after dose, with no evidence of major structural bias.

PopPK Model Evaluation

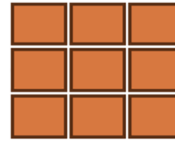
Across all age groups, the unified model demonstrated the most consistent performance, clinical accuracy, and nRMSE that were either superior to or statistically equivalent to the best-performing comparator in most subgroups. In neonates (see Figure 2a), the unified model was superior to or statistically equivalent to the best single model for accuracy, nRMSE, and *a priori* MPE, although the tobramycin model by Hennig et al¹³ showed better MPE and *a posteriori* MPE. The De Cock,¹⁶ Bijleveld,¹⁸ and Tong¹⁵ models also showed good performance in these metrics, though with slightly more bias. In children (see Figure 2b), the unified model demonstrated the highest accuracy and lowest nRMSE, but tended to over-predict levels (MPE: 20.1 (10.4–35.0)% *a priori*, 24.5 (13.4–37.4)% *a posteriori*). The models by Sherwin et al¹⁷ and Hennig et al¹³ showed the least bias *a priori* and *a posteriori*, respectively, and statistically equivalent nRMSE but worsened accuracy relative to the unified model. In adults (see Figure 2c), the unified model demonstrated the highest accuracy and lowest nRMSE, with a slight overprediction bias (14.4 (11.0–18.1)% *a priori*; 6.7 (4.3–9.0)% *a posteriori*) compared to the lower bias Hennig model (−8.1 (−11.0 to −5.2)% *a priori*; −5.8 (−8.1 to −3.7)% *a posteriori*).

Figure S11 shows the DV-PRED and DV-IPRED plots for each model. The unified model was also superior to or comparable to the best available single model by accuracy, MPE, and nRMSE across almost all subgroups defined by BMI (adults only), biological sex, and renal function (Figures S12–S17), and by trough and non-trough accuracy (Figure S18).

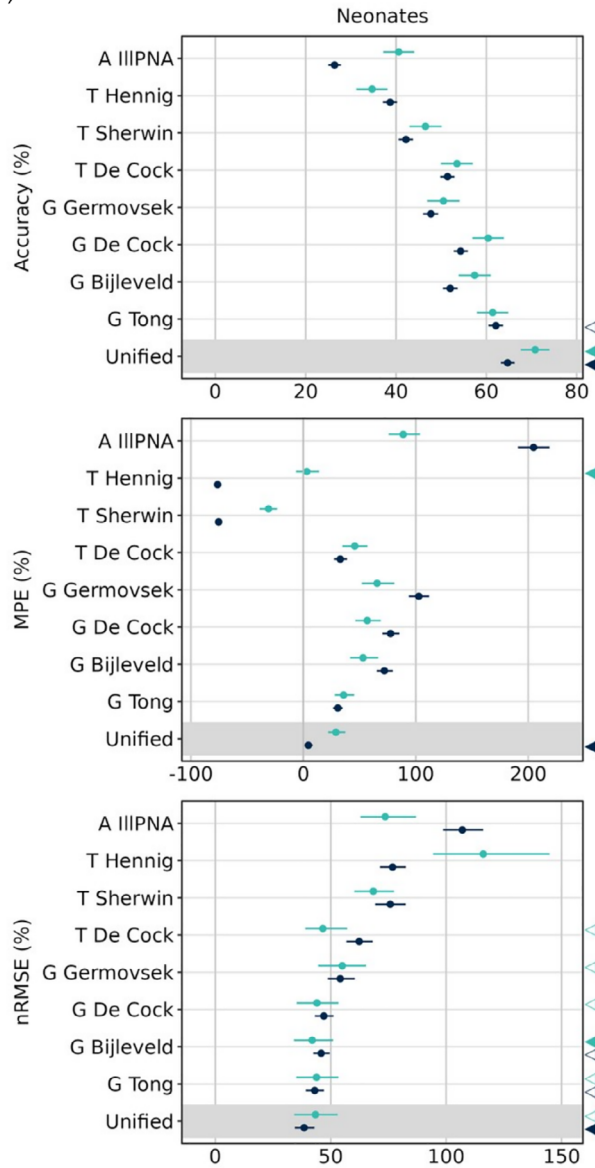
Model Robustness and Generalizability Evaluation

Leave-one-out model fits generally resulted in similar parameters (Figures S19 and S20), although parameter precision was diminished relative to the model fit with all available data. The model without adult tobramycin patients (N = 1013 patients removed) was the only one not identifiable, as indicated by a singular R matrix and failure of the covariance step in NONMEM. This remained true when removing the covariate effect for CF, which was informed only by patients in this dataset. Parameters describing age-related changes in renal function were the most sensitive to which populations were excluded. In particular, removing any gentamicin group significantly impacted the characterization of the age-clearance relationship (Figure S20), likely because gentamicin was the most widely represented drug with the most patients at either age extreme (adults = 1813; neonates = 1988). Removing pediatric gentamicin patients (N = 171) also had a large impact on the pediatric portion of the clearance-age curve, perhaps because they bridged the gap between the large adult and neonatal populations, providing information about variation across the lifespan. Removing

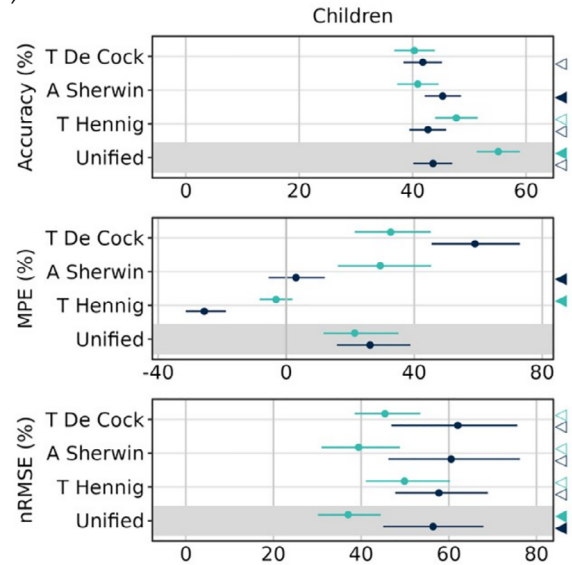
Unified : Evaluate on test data vs. published models



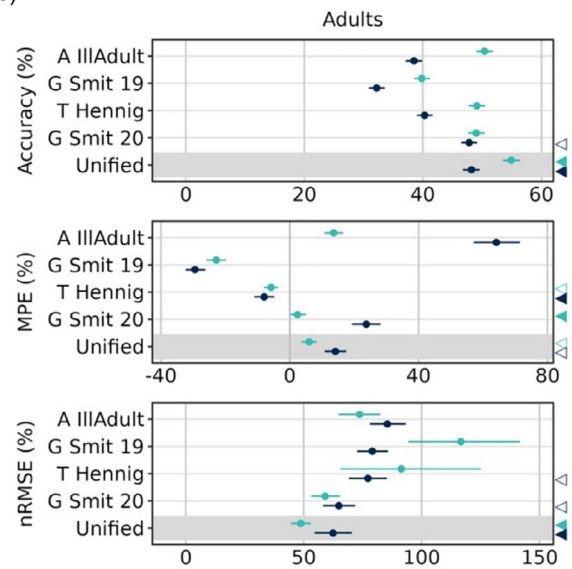
(a)



(b)



(c)



— apriori — aposteriori

Figure 2. Accuracy, mean percent error (MPE), and normalized root mean square error (nRMSE) of predictions for the newly fit model and published models on the test dataset for (a) neonatal patients ($N = 1153$ observations), (b) pediatric patients ($N = 406$), and (c) adult patients ($N = 2569$). The symbol $\blacktriangleleft$ indicates the best model; the symbol $\triangleleft$ indicates models that are not statistically distinguishable from the best model. For each model, the drug the model was developed for is given first (T = tobramycin, A = amikacin, and G = gentamicin), followed by the author name. IIIAdult and IIPNA refer to the Illamola model for adults and the Illamola model with postnatal age, respectively. Smit 19 and Smit 20 refer to reference [19] and Smit et al. 2020,³⁶ respectively.

Table 2. Parameter Estimates and Model Structure for the Newly Fitted Model.

Parameter	Description	Estimate	RSE (%)
$CL = \theta_{CL} \cdot \left(\frac{WT_{allo}}{70}\right)^{0.75} \cdot \left(\frac{F_{mat}}{F_{matR}}\right) \cdot \left(\frac{F_{age}}{F_{ageR}}\right) \cdot F_{CR} \cdot 1.17^{Tobra} \cdot 1.11^{Genta} \cdot 1.06^{CF}$ $F_{mat} = \frac{AGE^{2.61}}{AGE^{2.61} + 1.03^{2.61}}; F_{matR} = \frac{35^{2.61}}{35^{2.61} + 1.03^{2.61}}$ $F_{age} = \frac{AGE^{-Ge1a}}{AGE^{-Ge1a} + 66.1^{-6.1a}}; F_{ageR} = \frac{35^{-2eR1a}}{35^{-2eR1a} + 66.1^{-6.1a}}$ $F_{CR} = \exp(-0.631 \cdot (CR - CR_{std}))$			
θ_{CL} (L/h)	Drug clearance	5.1	2.3
	Tobramycin factor on CL	1.17	2.5
	Gentamicin factor on CL	1.11	2.3
	Cystic fibrosis factor on CL	1.06	3.8
	Serum creatinine exponent	0.63	11.2
	Maturation T50	1.03	59.3
	Maturation Hill exponent	2.61	2.9
	Age decay T50	66.1	0.5
	Age decay Hill exponent	-2.34	4.7
	Interindividual variability on CL	31.1	4.8
IIV on CL (%CV)			
$V_c = \theta_{Vc} \cdot \left(\frac{WT_{allo}}{70}\right)$			
θ_{Vc} (L)	Central volume of distribution	28.3	1.2
IIV on V_c (%CV)	Interindividual variability on V_c	21.2	13.8
$Q = \theta_Q \cdot \left(\frac{WT_{allo}}{70}\right)^{0.75} \cdot 0.84^{Ami}$			
θ_Q (L/h)	Intercompartmental clearance	1.45	4.2
	Amikacin factor on Q	0.84	8.4
$V_p = \theta_{Vp} \cdot \left(\frac{WT_{allo}}{70}\right) \cdot 1.41^{Ami}$			
θ_{Vp} (L)	Peripheral volume of distribution	24.7	4.5
	Amikacin factor on V_p	1.41	12.6
PROP TOBRA (%CV)	Proportional error for tobramycin	27.5	5.8
ADD TOBRA (mg/L)	Additive error for tobramycin	0.0648	75.3
PROP GENTA (%CV)	Proportional error for gentamicin	24.3	5.6
ADD GENTA (mg/L)	Additive error for gentamicin	0.137	10.7
PROP AMI (%CV)	Proportional error for amikacin	19.7	14.5
ADD AMI (mg/L)	Additive error for amikacin	0.827	21.4

adult gentamicin patients also substantially changed the intercompartmental clearance estimate from 1.45 to 1.56 L/h (Figure S21), suggesting data from adult patients also better inform drug distribution kinetics, or that the distribution component is stronger in adults. The predictive ability of these leave-one-out models to generalize to the unseen patient population is shown in Figure S22. The all-data model is generally best or tied with best across the board; however, for the leave-neonatal-gentamicin-out model, the performance is mixed.

Models fit on a single drug were estimable but with much lower precision for each parameter, or were not identifiable in the case of the amikacin-only model (Figure S23). The estimates for clearance from only tobramycin data (6.34 L/h, 24.8% higher than amikacin) or only gentamicin data (5.5 L/h, 8.3% higher than amikacin) models roughly match the expected values from the all-data model. The estimates on age-related effects on clearance were less precise, particularly on renal maturation, and resulted in slightly different mathematical relationships between clearance and age (Figure S24), with the amikacin-only model resulting in a sharper peak during the pediatric phase, and the

tobramycin-only model resulting in a slightly more rapid decline in clearance after age 35.

Predictive performance of single-drug models on the drugs they were trained on (i.e., the amikacin-only model being evaluated on amikacin test data) was statistically tied with the all-data model (Figure 3). However, when evaluated on the non-training drugs (i.e., the amikacin-only model evaluated on tobramycin and gentamicin test data), the value of training across all drugs became apparent. Though the single-drug models were statistically equivalent to the all-data model across many metrics, the all-data model was consistently statistically equivalent and sometimes superior in bias or nRMSE (Figure 4). This superiority was especially evident when comparing MPE on the all-data model to the amikacin-only model on non-amikacin data in neonates and adults, and similarly for the tobramycin-only model on non-tobramycin data in neonates and adults.

Discussion

Model-informed precision dosing depends on popPK models that are developed in sufficiently representative

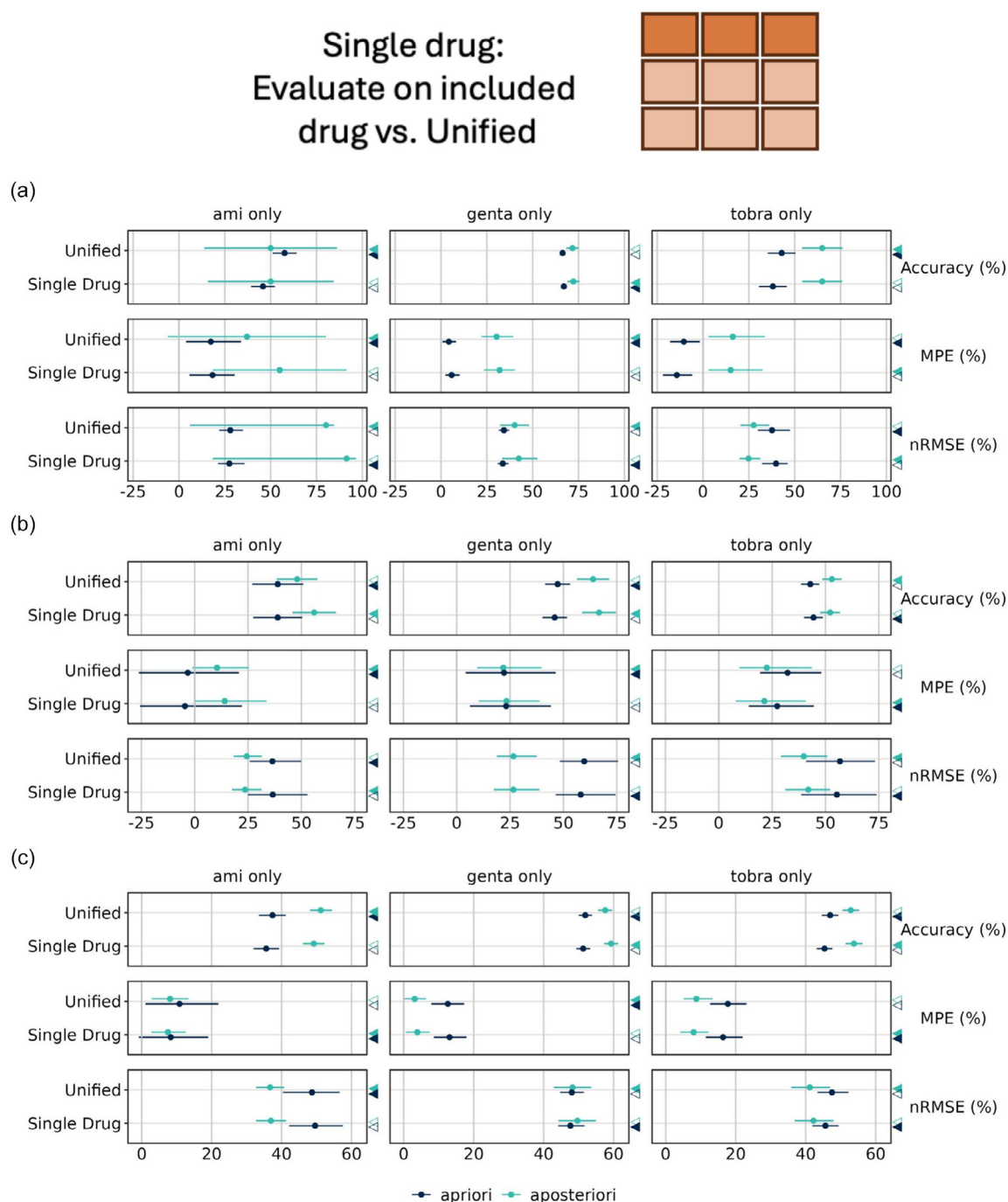


Figure 3. Accuracy, mean percent error (MPE), and normalized root mean square error (nRMSE) of predictions for the newly fit model and the same model trained only on one drug, evaluated on only the included drug in the test dataset, in (a) neonates, (b) children, and (c) adults. For example, the single drug model trained with amikacin (“Single Drug”) is evaluated against the model with all training data (“Unified”) in the leftmost column (“ami only”). The symbol ◀ indicates the best model; the symbol ◁ indicates models that are not statistically distinguishable from the best model.

patient populations to support accurate drug exposure predictions and guide dosing decisions in unseen populations. In this study, we developed a unified popPK model using nearly 6000 treatment courses spanning three aminoglycosides and all age groups and demonstrated that this model predicted as well as or better

than existing published models. To our knowledge, this is the first popPK model to formally test and quantify shared pharmacokinetic behavior across aminoglycosides using a single all-ages dataset, and to systematically assess the impact of including only individual drugs or excluding drug-age subgroups on parameter

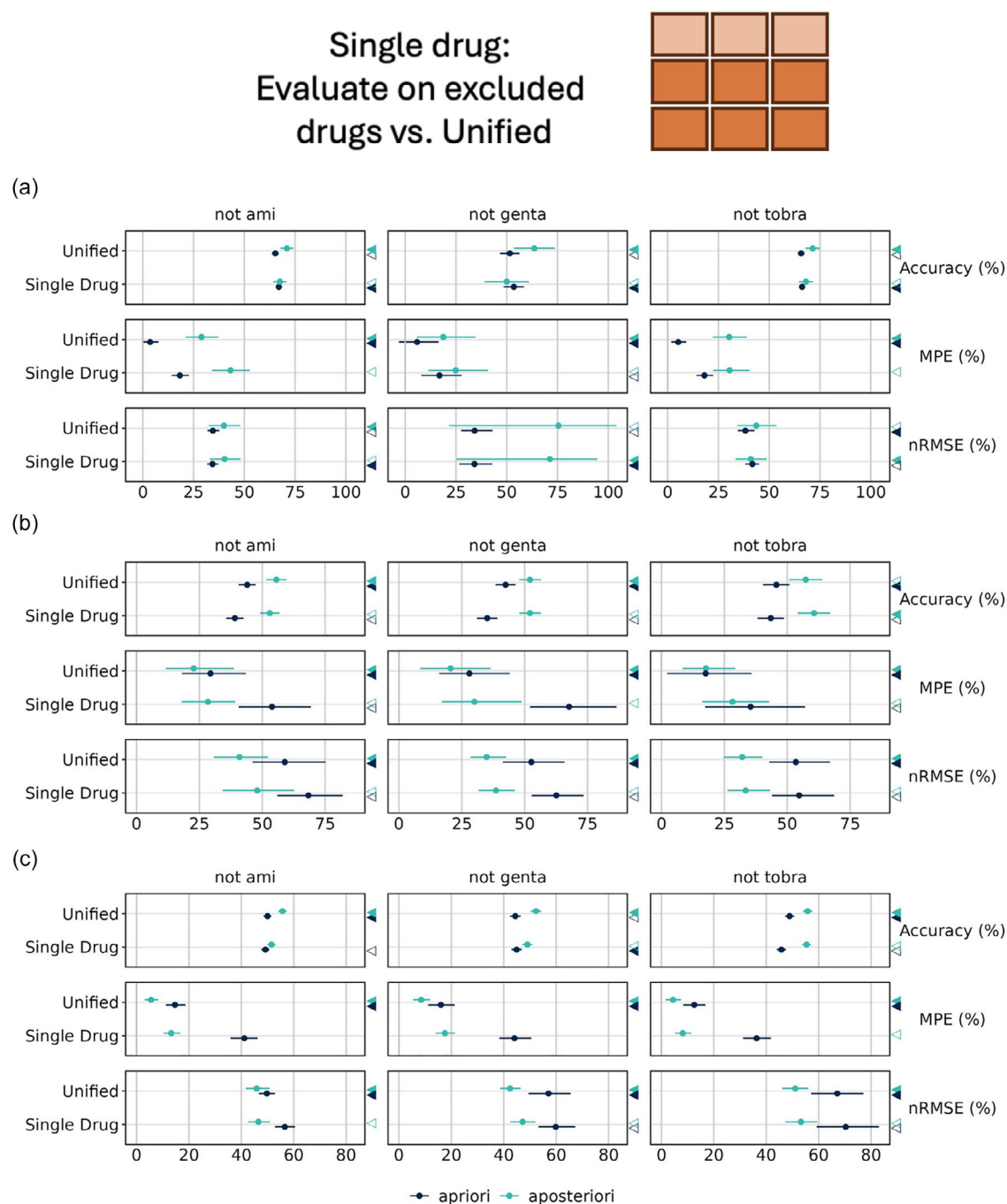


Figure 4. Accuracy, mean percent error (MPE), and normalized root mean square error (nRMSE) of predictions for the newly fit model and the same model trained only on one drug, evaluated on test data not from that drug, in (a) neonates, (b) children, and (c) adults. For example, the single drug model trained with amikacin ("Single Drug") is evaluated against the model with all training data ("Unified") in the leftmost column ("not ami") on test data from gentamicin and tobramycin. The symbol ◀ indicates the best model; the symbol ◁ indicates models that are not statistically distinguishable from the best model.

estimates and predictive performance. These findings support the feasibility of unified modeling approaches across drug families with similar PK, and suggest that leveraging data from other drugs with similar PK may improve model predictive performance across diverse patient populations.

The unified model demonstrated robust performance across clinically relevant patient subgroups, including those with extreme obesity and severely impaired renal function, which are often considered challenging patient subgroups to accurately dose in routine clinical care. Predictive performance was superior

to or comparable to the best available published model across almost all subgroup categories, though *a posteriori* performance was slightly worse than *a priori* in some renal subgroups, likely reflecting the small sample sizes. These findings support the use of the unified model across the full spectrum of patients encountered in routine clinical care and also support the value of developing popPK models on broad, representative datasets rather than narrow subpopulations, as a model trained on diverse patient data generalizes better to the complex patients most likely to benefit from precision dosing. Although body size-dependent biological differences in renal elimination between aminoglycosides may exist,¹⁹ we found no evidence of model misspecification across the full BMI range for any of the three drugs, suggesting that the covariate structure adequately captured obesity-related effects on clearance empirically.

Prior work has demonstrated the feasibility and value of all-age modeling in related contexts. For example, Holford et al demonstrated that related antimicrobials can be modeled using shared physiological components while retaining drug-specific parameters,²⁵ and Davda et al explored cross-monoclonal antibody modeling strategies.³⁷ Examples of popPK models for single drugs across the human lifespan^{13,28} and maturation functions for neonates and children^{14,38} also show that this approach has been evaluated in smaller datasets. In contrast, this study jointly estimates shared PK parameters across three aminoglycosides and the full age spectrum, while retaining drug-specific effects supported by the data, and evaluates the generalizability and robustness of this approach. This unified framework demonstrates that a single model can capture both shared pharmacokinetic structure and meaningful drug-specific differences.

Pooling data across aminoglycosides and age groups improved the stability and generalizability of parameter estimation, allowing common parameters like renal maturation, age-related decline in clearance, and inter-individual variability to be estimated using substantially more information than was available in any individual drug dataset. This was particularly important for nonlinear age-related effects, as young patients were underrepresented in the datasets available for tobramycin and amikacin. As a result, the unified model produced more stable and precise estimates of age-related clearance changes across the lifespan, which supports more reliable dosing recommendations in populations where individual drugs have historically lacked sufficient developmental data.³⁹

Some differences in PK were observed between drugs, as indicated by the improved model fits obtained after including drug-specific covariates. At the same time, pooling datasets from several drugs allowed precise estimation of PK parameters and variability terms,

including shared inter-individual variability in clearance and central volume of distribution. In contrast, single-drug models relied on smaller datasets, reducing precision of parameter estimates. The amikacin-only model, developed on the smallest cohort, was not identifiable when modeled independently, further supporting the value of a unified approach. These findings support a shared structural pharmacokinetic framework with limited drug-specific adjustments.

This unified model may also perform reasonably well when extrapolated to an unmodeled aminoglycoside (e.g., kanamycin), given the shared PK characteristics within the aminoglycoside class. However, selective modifications of individual PK parameters may still be necessary, as observed for intercompartmental clearance (Q) and peripheral volume of distribution (V2) in amikacin. Under the assumption that age-related changes in clearance are constant across aminoglycosides, data from a single age group may be adequate to support construction of a popPK model applicable across the full age spectrum.

Leave-one-out analyses highlighted the statistical importance of specific subpopulations for estimating age-dependent effects. Neonatal gentamicin data constituted the largest neonatal subgroup, and exclusion of these patients produced substantially different estimates of renal maturation parameters. Similarly, pediatric gentamicin patients were a critical bridge between large neonatal and adult cohorts; their exclusion resulted in exaggerated clearance estimates for pediatric patients.

Despite these sensitivities, most leave-one-out models remained identifiable and demonstrated predictive performance comparable to the full model when evaluated on unseen patient populations. These findings indicate that the unified model is generally robust to exclusion of individual drug-age group combinations while still benefiting from broad age and drug coverage. In applications where data for a specific subgroup are unavailable (e.g., neonates for a given drug), this approach supports transferring knowledge from similar drugs and age-matched populations, providing a potential strategy for model-informed drug development or MIPD in data-limited cases.

Several design decisions contributed to the robustness and generalizability of the unified aminoglycoside model. Body size was incorporated using FFM down to 3 years of age,³¹ with linear interpolation between weight and FFM from 1 to 3 years, and body weight below 1 year of age. This enabled allometric scaling across the full age spectrum, allowing size-related variability to be separated from age- and renal-related effects. More recently, methods to describe changes in FFM across the lifespan have been published, although these equations are relatively complex and our simpler approach

allowed for unbiased parameter estimates. Similarly, we used age- and sex-adjusted typical serum creatinine for normalizing patient serum creatinine, allowing expression of renal function across the whole age spectrum. Alternatives like Cockcroft–Gault,³⁵ CKD-EPI,⁴⁰ and the revised Schwartz bedside equation⁴¹ were developed on patient populations of narrower age ranges, limiting their applicability in a unified model across the lifespan.

This study has several limitations. First, the use of routine clinical care data from multiple institutions may introduce heterogeneity in dosing, sampling, and assay practice. Second, sampling density per patient was low, reflecting real-world practice, which limited our ability to estimate distribution parameters, particularly inter-compartmental clearance and peripheral volume of distribution.^{42,43} This low sampling density also limited our ability to assess accuracy of area under the curve (AUC) estimation, which has been reported as a predictor of toxicity and efficacy.^{44,45} Third, this model was developed exclusively on United States-based patients, potentially limiting generalizability to other settings. Additionally, several clinically important subgroups, including critically ill patients, very elderly adults, patients with renal impairment, and cystic fibrosis patients, were relatively small. For critically ill patients, this precluded subgroup-specific parameter estimation, while for the remaining groups, it limited the precision of the estimated covariate relationships. Fourth, due to data availability, we could not assess model performance on other aminoglycosides. However, the results presented here provide a roadmap for extending these findings to other drugs within the class. Finally, the model assumes shared PK parameters and variability between aminoglycosides. This is supported by prior work and by the data but warrants prospective validation.

Future work could extend this model by incorporating more densely sampled PK data to improve the identifiability of parameters and reduce uncertainty in age-related effects. Focused refits in populations of interest, like critically ill patients, may further improve predictive performance for those vulnerable populations. Prospective evaluation of this model in routine clinical care would also be valuable in assessing its impact on dosing decisions and patient outcomes. Models could also be used to estimate AUC, which is increasingly being used as an efficacy metric for aminoglycosides. Assessing the agreement of these estimates across models and linking them to clinical outcomes would substantially strengthen the case for AUC-based dosing. These efforts would further refine and validate this unified model for use in clinical model-informed precision dosing across aminoglycosides and age groups. Application of the methodology of unified drug-class modeling demonstrated here could be evaluated in other drug classes as well.

Conclusion

In this study, we developed and evaluated a unified popPK model for aminoglycosides spanning neonatal, pediatric, and adult patients using a large, real-world clinical dataset. By leveraging shared physiological and pharmacokinetic parameters while incorporating targeted drug-specific differences, the model outperformed existing models across all aminoglycosides and age groups. Robustness and generalizability analyses suggested that the model was resilient to the exclusion of individual drug–age group combinations, and generalized better to unseen drugs compared to single drug models. These findings support the use of a unified aminoglycoside model for Bayesian model-informed precision dosing and demonstrate that pooling data across related drugs and age groups can enhance model performance in routine clinical practice.

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Author Contributions

Dominic Tong: Conceptualization; methodology; software; formal analysis; data curation; visualization; writing—original draft; writing—review and editing. Jon Faldasz: Conceptualization; methodology; writing—review and editing. Ron Keizer: Conceptualization; methodology; writing—review and editing. Jasmine Hughes: Conceptualization; methodology; visualization; writing—original draft; writing—review and editing.

Conflicts of Interest

D.M.H.T., J.D.F., R.J.K., and J.H.H. are employees and stockholders of Insight Rx, Inc.

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Data Availability Statement

Due to its proprietary nature, supporting data cannot be made available.

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